2. Review: Fundamental matrices

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Lecture due Mar 6, 2024 20:30 IST

Fundamental matrix review 1

1/1 point (graded)

Assume that in each matrix, the columns are solutions to one system of ODEs.

Which of the following matrices could be fundamental matrices?

- [x] $\begin{pmatrix} \cos(t) & \sin(t) \\ -\sin(t) & \cos(t) \end{pmatrix}$
- [ ] $\begin{pmatrix} e^t & e^{2t} \\ -2e^t & -2e^{2t} \end{pmatrix}$
- [x] $\begin{pmatrix} e^{-5t} & e^t & e^t \\ 0 & e^t & 0 \\ 0 & 0 & e^t \end{pmatrix}$
- [x] $\begin{pmatrix} e^{it} & e^{-it} \\ -ie^{it} & ie^{-it} \end{pmatrix}$

Solution:

We only need to check that $\mathbf{X}(0)$ is invertible! Let us first proceed in order of the choices:

- $\left.\begin{pmatrix} \cos(t) & \sin(t) \\ -\sin(t) & \cos(t) \end{pmatrix}\right|_{t=0} = I$ is invertible
- $\left.\begin{pmatrix} e^t & e^{2t} \\ -2e^t & -2e^{2t} \end{pmatrix}\right|_{t=0} = \begin{pmatrix} 1 & 1 \\ -2 & -2 \end{pmatrix}$ is singular.
- $\left.\begin{pmatrix} e^{-5t} & e^t & e^t \\ 0 & e^t & 0 \\ 0 & 0 & e^t \end{pmatrix}\right|_{t=0} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is invertible.
- $\left.\begin{pmatrix} e^{it} & e^{-it} \\ -ie^{it} & ie^{-it} \end{pmatrix}\right|_{t=0} = \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix}$ is invertible.

Any of the invertible matrices above could be a fundamental matrix given an appropriate systems of ODEs.

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You have used 1 of 3 attempts

Answers are displayed within the problem

Fundamental matrix review 2

1/1 point (graded)

We also use the terms **fundamental matrix of a single higher order ODE** to mean a fundamental matrix of its companion system.

Determine which of the following second order ODEs or system of ODEs has the following fundamental matrix:

$$\begin{pmatrix} \cos(t) & \sin(t) \\ -\sin(t) & \cos(t) \end{pmatrix}$$

Calculator

- [x] $\ddot{x} + x = 0$
- [ ] $\ddot{x} + \dot{x} + x = 0$
- [x] $\dot{\mathbf{x}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \mathbf{x}$
- [ ] $\dot{\mathbf{x}} = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} \mathbf{x}$

✔

Solution:

Each system is the companion matrix of one of the second order ODEs, so we only need to check the systems. Inspection shows that

$$\frac{d}{dt}\begin{pmatrix} \cos(t) & \sin(t) \\ -\sin(t) & \cos(t) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} \cos(t) & \sin(t) \\ -\sin(t) & \cos(t) \end{pmatrix}$$

Thus both $\ddot{x} + x = 0$ and its companion matrix $\dot{\mathbf{x}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \mathbf{x}$ have a fundamental matrix $\begin{pmatrix} \cos(t) & \sin(t) \\ -\sin(t) & \cos(t) \end{pmatrix}$.

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2. Review: Fundamental matrices

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